Politecnico di Milano

Biochip A.Y. 2019/2020 - M. Carminati

On-Line Exam of 17.06.2020

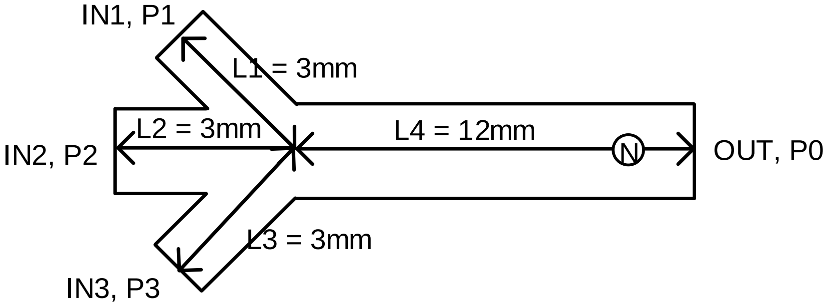
Test duration: 2 hours – No support material admitted.

All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.

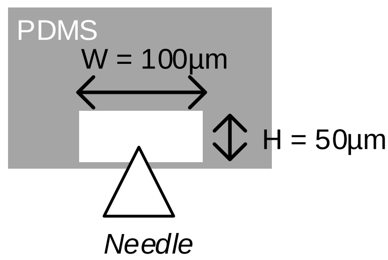
The exam focuses on an electrochemical glucose sensor based on a micro-fabricated epidermal needle electrode.

1. Before use on the patient, the needle is tested in a simple microfluidic chip (shown in the figure) made of PDMS. In all sections the cross section is the same (W = 100μm, H = 50μm). Assuming now that inlet 2 is closed, P1 = 3atm , P3 = 2atm, find the velocity of the fluid at point N (P0 = 1atm = 101 kPa, $\eta_{water} = 10^{-3}$ Pa·s).

Top View

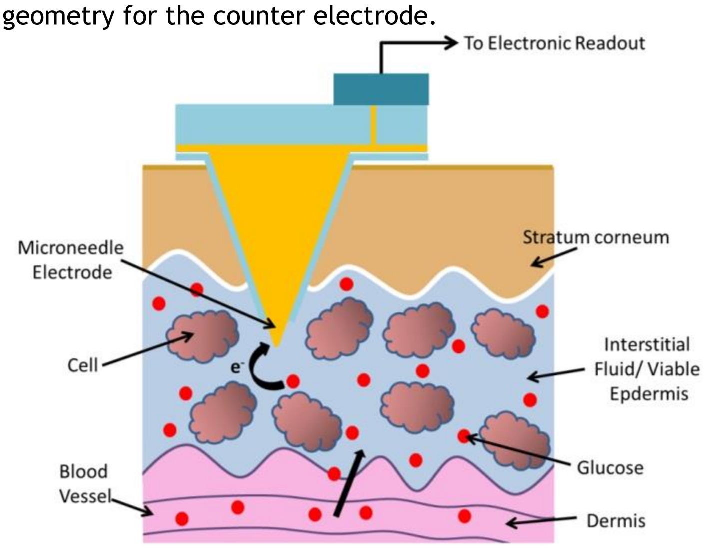


Channel Cross Section

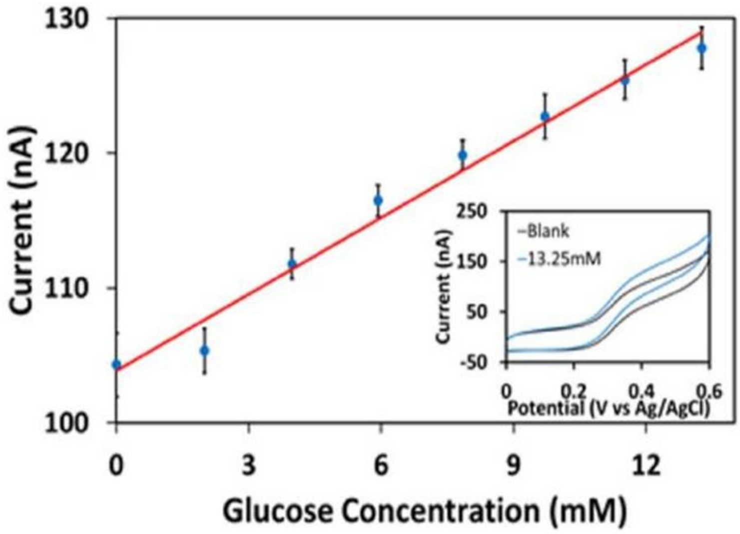


2. How does temperature affect the answer to the previous question (1)?
3. What is the impact of the bottom thin flexible membrane on the electrical equivalent model of the fluidic circuit?
4. Now consider all three inlets working, if Q2 = 10μL/s, find Q1 = Q3 in order to confine the middle fluid in a virtual channel of width 20μm.
5. Assume that the three flow rates of question (4) are generated by peristaltic pumps. Propose a solution to avoid that the ripple of pumps alters the width of the focused channel.
6. Discuss for both syringe pumps and peristaltic pumps if they can operate in both directions (push and withdrawal). From the point of view of air bubbles, is it better to operate in push or aspiration?

7. Now the micro-needle (working electrode) is inserted in the skin of the patient to measure in real-time the concentration of glucose in the interstitial fluid. Briefly describe the working principle of an electrochemical glucose sensor and suggest a geometry for the counter electrode.



8. Assuming that the needle is coated with gold and its area is 500μm^2, assuming that the interstitial fluid has the same properties of PBS (C_{DL}' = 0.1 pF/μm^2, conductivity 1.5S/m), find the small-signal impedance equivalent of the electrode and explain which parameters can be estimated.
9. Considering the detector response of the plot, assuming a max. concentration of glucose of 10mM and a single power supply voltage of 10V, size the value of the feedback resistor of the transimpedance amplifier.



10. Find the bandwidth of the amplifier.
11. Considering the same detector response and a total input referred noise (white) of 50fA/sqrt(Hz), find the limit of detection of the sensor.
12. Describe the fabrication steps of the metal needle fabricated on polymer cones, as illustrated in the figure.